

7.Demonstration of Activation Functions (ReLU, Sigmoid, and Tanh) Using Synthetic Data

Introduction

Activation functions are mathematical functions used in Artificial Neural Networks (ANNs) to determine the output of a neuron. They introduce **non-linearity**, enabling neural networks to learn complex patterns and relationships in data.

Without activation functions, a neural network behaves like a linear model and cannot solve complex machine learning problems.

Why Activation Functions are Important?

- Introduce non-linearity into neural networks.
- Help learn complex patterns from data.
- Improve prediction accuracy.
- Enable deep learning models to solve classification and regression problems.
- Control the output of neurons.

Sigmoid Activation Function

The Sigmoid function maps the input values into the range **0 to 1**. It is commonly used in the output layer of binary classification problems.

Formula

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Characteristics

- Output Range: 0 to 1
- Smooth S-shaped curve
- Suitable for binary classification

Tanh Activation Function

The Tanh (Hyperbolic Tangent) function maps input values between **-1 and 1**. It is zero-centered and often performs better than the Sigmoid function in hidden layers.

Formula

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Characteristics

- Output Range: -1 to 1
- Zero-centered
- Used in hidden layers

ReLU Activation Function

ReLU (Rectified Linear Unit) is the most widely used activation function in deep learning. It outputs zero for negative values and returns the input value for positive values.

Formula

[$\text{ReLU}(x)=\max(0,x)$]

Characteristics

- Output Range: 0 to ∞
- Fast and computationally efficient
- Commonly used in hidden layers of deep neural networks

Comparison of Activation Functions

Activation Function	Output Range	Common Use
Sigmoid	0 to 1	Binary Classification
Tanh	-1 to 1	Hidden Layers
ReLU	0 to ∞	Deep Learning Models

Algorithm

1. Import the required libraries.
2. Generate synthetic input data using NumPy.
3. Define the Sigmoid activation function.
4. Define the Tanh activation function.
5. Define the ReLU activation function.
6. Apply each activation function to the input data.
7. Plot the output curves using Matplotlib.
8. Compare the behavior of all three activation functions.

```
In [1]: # Import Required Libraries

import numpy as np
import matplotlib.pyplot as plt

# Generate Synthetic Data

x = np.linspace(-10, 10, 200)

# Define Activation Functions

def sigmoid(x):
    return 1 / (1 + np.exp(-x))

def tanh(x):
    return np.tanh(x)

def relu(x):
    return np.maximum(0, x)

# Apply Activation Functions

sigmoid_output = sigmoid(x)
tanh_output = tanh(x)
relu_output = relu(x)

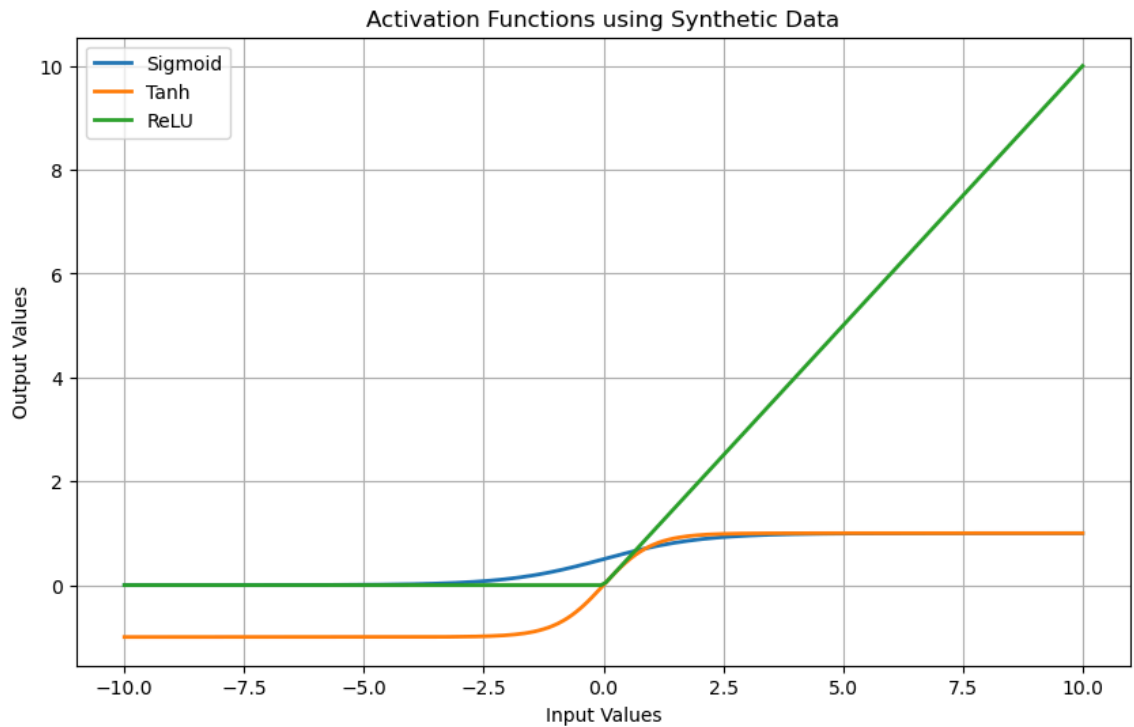
# Plot the Activation Functions

plt.figure(figsize=(10, 6))

plt.plot(x, sigmoid_output, label='Sigmoid', linewidth=2)
plt.plot(x, tanh_output, label='Tanh', linewidth=2)
plt.plot(x, relu_output, label='ReLU', linewidth=2)

plt.title("Activation Functions using Synthetic Data")
plt.xlabel("Input Values")
plt.ylabel("Output Values")
plt.grid(True)
plt.legend()

plt.show()
```



Observation

- Sigmoid produces an S-shaped curve with outputs between **0 and 1**.
- Tanh produces outputs between **-1 and 1**, making it zero-centered.
- ReLU outputs **0** for negative inputs and increases linearly for positive inputs.
- ReLU is the most commonly used activation function in modern deep learning models.

Result

The working of Sigmoid, Tanh, and ReLU activation functions was successfully demonstrated using synthetic data. The plotted graphs clearly illustrate the behavior and output range of each activation function.